

QUESTIONS 31 / 31 completed

Arithmetic **ARITHMETIC**

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java 

Run

Save

```
1 import java.util.*;
2 public class arithmetic{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int a=sc.nextInt(),b=sc.nextInt();
6         System.out.println("Sum:" +(a+b));
7         System.out.println("Differnce:"+(a-b));
8         System.out.println("Product:"+(a*b));
9         System.out.println("Quotient:"+(a/b));
10    }
11 }
```

OUTPUT

```
Sum:15
Differnce:5
Product:50
Quotient:2
```

INPUT

```
10
5
```

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class maxthree{
3     public static void main(String[] args){
4         Scanner sc = new Scanner (System.in);
5         int a = sc.nextInt();
6         int b = sc.nextInt();
7         int c = sc.nextInt();
8         int max;
9         if(a>=b && a>=c){
10             max=a;
11         }
12         else if(b>=a && b>=c){
13             max=b;
14         }
15         else{
16             max=c;
17         }
18         System.out.println("Maximum number is "+max);
19     }
20 }
```

OUTPUT

```
Maximum number is 6
```

INPUT

```
3
6
1
```

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class leap{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int y = sc.nextInt();
6         if(y%4==0 && y%100!=0||(y%400==0)){
7             System.out.println("leap year");
8         }
9         else{
10            System.out.println("Not a leap year");
11        }
12    }
13 }
```

OUTPUT

Not a leap year

INPUT

2014

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class fibb{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int a=0;
7         int b=1;
8         System.out.print("Fibonacci series up to "+n+": ");
9         while(a<=n){
10             System.out.print(a+" ");
11             int c = a+b;
12             a=b;
13             b=c;
14         }
15     }
16 }
```

INPUT

10

OUTPUT

Fibonacci series up to 10: 0 1 1 2 3 5 8

QUESTIONS 31 / 31 completed

Factorial **FACTORIAL**

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java 

Run

Save

```
1 import java.util.Scanner;
2 public class factorial{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n= sc.nextInt();
6         long fact=1;
7         if(n<0){
8             System.out.println("Entered number in negative");
9         }
10        else{
11            for(int i=1;i<=n;i++){
12                fact*=i;
13            }
14            System.out.println(fact);
15        }
16    }
17 }
```

OUTPUT

120

INPUT

5

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

```
**
***
****
*****
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class rt{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for(int i=1;i<=n;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print("**");
9             }
10            System.out.println();
11        }
12    }
13 }
```

INPUT

5

OUTPUT

```
*
**
***
****
*****
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class search{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int a[]=new int[n];
7         for(int i =0;i<n;i++)
8             a[i]=sc.nextInt();
9         int key = sc.nextInt();
10        int l=0,h=n-1,f=0;
11        while(l<=h){
12            int m=(l+h)/2;
13            if(a[m]==key){
14                System.out.println("Element found at position "+(m+1));
15                f=1;
16                break;
17            }
18            else if(a[m]<key)
19                l=m+1;
20        }
21        if(f==0)
```

OUTPUT

Element found at position 3

INPUT

```
5
10 20 30 40 50
30
```

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int a =sc.nextInt();
6         int b =sc.nextInt();
7         while(b!=0){
8             int t=b;
9             b=a%b;
10            a=t;
11        }
12        System.out.println(a);
13
14    }
15 }
```

INPUT

12
18

OUTPUT

6

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n =sc.nextInt();
6         int f = 0;
7         if(n<=1)
8             f=1;
9         for(int i=2;i<=n/2;i++){
10             if(n%i==0){
11                 f=1;
12                 break;
13             }
14         }
15         if(f==0)
16             System.out.println("Prime");
17         else{
18             System.out.println("Not prime");
19         }
20     }
21 }
```

OUTPUT

Prime

INPUT

17

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int a[]=new int[n];
7         for(int i=0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        for(int i=0;i<n-1;i++){
11            int min=i;
12            for(int j=i+1;j<n;j++){
13                if(a[j]<a[min])
14                    min=j;
15            }
16            int t=a[i];
17            a[i]=a[min];
18            a[min]=t;
19        }
20        for(int i=0;i<n;i++){
21            System.out.println(a[i]+" ");
22        }
23    }
24 }
```

OUTPUT

```
11
12
22
25
64
```

INPUT

```
5
64 25 12 22 11
```

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class tab{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for(int i=1;i<=10;i++){
7             System.out.println(n + "*" + i + "=" + (n*i));
8         }
9     }
10 }
11 }
```

OUTPUT

```
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50
```

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     static int fact(int n){
4         if(n==0 || n==1)
5             return 1;
6         return n*fact(n-1);
7     }
8     public static void main(String[] args){
9         Scanner sc = new Scanner(System.in);
10        int n=sc.nextInt();
11        System.out.println(fact(n));
12    }
13 }
```

OUTPUT

120

INPUT

5

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

```
*****
****
***
**
*
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class pt{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for(int i=n;i>=1;i--){
7             for(int j=1;j<=i;j++){
8                 System.out.print("*");
9             }
10            System.out.println();
11        }
12    }
13 }
```

INPUT

5

OUTPUT

```
*****
****
***
**
*
```

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int a=sc.nextInt(),b=sc.nextInt();
6         int l=Math.max(a,b);
7         while(true){
8             if(l%a==0 && l%b==0){
9                 System.out.println(l);
10                break;
11            }
12            l++;
13        }
14    }
15 }
```

OUTPUT

36

INPUT

12
18

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class dtob{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         String bs=Integer.toBinaryString(n);
7         System.out.println(bs);
8     }
9 }
```

OUTPUT

1100

INPUT

12

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class btod{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         String b = sc.nextLine();
6         int d=Integer.parseInt(b,2);
7         System.out.println(d);
8     }
9 }
10 }
```

OUTPUT

13

INPUT

1101

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class sum{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=sc.nextInt();
6         int s=0;
7         while(n!=0){
8             int dig = n%10;
9             s=s+dig;
10            n=n/10;
11        }
12        System.out.println(s);
13    }
14 }
```

INPUT

124

OUTPUT

7

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class prod{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int p = 1;
7         while(n!=0){
8             int d=n%10;
9             p=p*d;
10            n=n/10;
11        }
12        System.out.println("Product of the digits is "+p);
13    }
14 }
```

OUTPUT

Product of the digits is 6

INPUT

123

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int b=sc.nextInt();
6         int e=sc.nextInt();
7         int p=1;
8         for(int i=1;i<=e;i++){
9             p=p*b;
10        }
11        System.out.println(p);
12    }
13 }
```

INPUT

```
2
5
```

OUTPUT

```
32
```

QUESTIONS 31 / 31 completed

Area Circle ▾

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 public class area{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int r = sc.nextInt();
6         System.out.println("Area of the circle is "+(3.14*r*r));
7     }
8 }
```

INPUT

5

OUTPUT

Area of the circle is 78.5

QUESTIONS 31 / 31 completed

Shpere **SHPERE**

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java 

Run

Save

```
1 import java.util.Scanner;
2 public class shpere{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int r=sc.nextInt();
6         System.out.println((4/3)*3.14*r*r*r);
7     }
8 }
```

INPUT

5

OUTPUT

System Error: Could not reach compiler or server error occurred.

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class palin{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         String s=sc.nextLine();
6         String rev="";
7         for(int i =s.length()-1;i>=0;i--){
8             rev += s.charAt(i);
9         }
10        if(s.equals(rev))
11            System.out.println("Palindrome");
12        else
13            System.out.println("Not palindrome");
14    }
15 }
```

OUTPUT

Palindrome

INPUT

madam

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class armstrong{
3     public static void main(String[] args){
4         Scanner sc = new Scanner (System.in);
5         int n=sc.nextInt();
6         int sum=0;
7         int o = n;
8         while(n>0){
9             int d = n%10;
10            sum+= d*d*d;
11            n=n/10;
12        }
13        if(sum==o){
14            System.out.println("Amstrong number");
15        }
16        else{
17            System.out.println("Not an amstrong number");
18        }
19    }
20 }
21 }
```

OUTPUT

Amstrong number

INPUT

153

QUESTIONS 31 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to
print the following
pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class npt{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for(int i=1;i<=n;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print(i);
9             }
10            System.out.println();
11        }
12    }
13 }
```

INPUT

5

OUTPUT

```
1
22
333
4444
55555
```


QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class perfect{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int sum=0;
7         for(int i=1;i<n;i++){
8             if(n%i==0)
9                 sum += i;
10        }
11        if(sum==n)
12            System.out.println("Perfect number");
13        else
14            System.out.println("Not perfect number");
15    }
16 }
```

OUTPUT

Perfect number

INPUT

28

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int[] a =new int[n];
7         for(int i =0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int max=a[0];
11        for(int i=1;i<n;i++){
12            if(a[i]>max)
13                max=a[i];
14        }
15        System.out.println(max);
16    }}
17 }}
```

OUTPUT

40

INPUT

5
10 25 7 40 18

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[]a=new int[n];
7         for(int i=0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int max=a[0],s=a[0];
11        for(int i=1;i<n;i++){
12            if(a[i]>max){
13                s=max;
14                max=a[i];
15            }
16            else if(a[i]>s&&a[i]!=max){
17                s=a[i];
18            }
19        }
20        System.out.println(s);
21    }
```

INPUT

5
10
25
40
18
30

OUTPUT

30

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=sc.nextInt();
6         int[] a=new int[n];
7         for(int i =0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int min =a[0];
11        for(int i=1;i<n;i++){
12            if(a[i]<min)
13                min=a[i];
14        }
15        System.out.println(min);
16    }
17 }
```

OUTPUT

3

INPUT

```
5
12
8
25
3
10
```

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=sc.nextInt();
6         int[] a=new int[n];
7         for(int i=0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int min =Integer.MAX_VALUE;
11        int s=Integer.MAX_VALUE;
12        for(int i =0;i<n;i++){
13            if(a[i]<min){
14                s=min;
15                min=a[i];
16            }
17            else if(a[i]<s&&a[i]!=min)
18                s=a[i];
19        }
20        System.out.println(s);
21    }
```

OUTPUT

5

INPUT

5
10
5
20
3
8

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for(int i=0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        for(int i=0;i<n-1;i++){
11            for(int j=0;j<n-i-1;j++){
12                if(a[j]>a[j+1]){
13                    int t =a[j];
14                    a[j]=a[j+1];
15                    a[j+1]=t;
16                }
17            }
18        }
19        for(int i =0;i<n;i++)
20            System.out.println(a[i]+" ");
21    }
```

OUTPUT

```
1
2
3
5
8
```

INPUT

```
5
5
3
8
1
2
```

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for(int i=0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        for(int i=1;i<n;i++){
11            int key=a[i];
12            int j=i-1;
13            while(j>=0 && a[j]>key){
14                a[j+1]=a[j];
15                j--;
16            }
17            a[j+1]=key;
18        }
19        for(int i =0;i<n;i++)
20            System.out.println(a[i]+" ");
21    }
```

OUTPUT

```
1
2
3
5
8
```

INPUT

```
5
5
3
8
1
2
```